CAGE-Fake: Article Trustworthiness Audit Report
Calibration-Aware Gating & Evidence-Grounded Reasoning for Trustworthy Fake News Triage

AUDIT ID: AUD-123818
DATE: 20261027_232646
FRAMEWORK: CAGE-Fake v2.0

LEGAL & FACT-CHECKING POLICY: A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify factual reality. URL extraction, pattern recognition, and external evidence verification remain strictly separate processes. The pipeline abstains (**UNCERTAIN**) whenever evidence is missing, conflicting, or model consensus is low.

Title: Man gets life term for abducting, raping minor
Source: Manual Text Submission
Word Count: 179 words | **Extraction:** Direct Manual Input
Triage Policy: Evidence-Grounded Three-Way Gating

TRIAGE VERDICT: UNCERTAIN
Calibrated Confidence: 95.9% (τ = 65.0%)
Model Disagreement (std): 0.092 (δ = 0.20)
Safety Trigger: Missing Corroborating Evidence

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Vote	Confidence
Random Forest	70.6%	FAKE	70.6%
XGBoost	90.4%	FAKE	90.4%
LightGBM	93.5%	FAKE	93.5%
Logistic Regression	91.3%	FAKE	91.3%
CAGE-Fake Stack	95.9%	FAKE	95.9%

Consensus Status: Strong Consensus
Vote Distribution: **5 FAKE** vs **0 REAL**
Base Disagreement (4-base std): 0.092 (δ = 0.20)
5-Model Consensus Spread: 0.091
[======----------]
Safety Assessment: Models strongly agree on the verdict with minor variance.

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	89.5/100	Excellent / Restrained
Source Attribution	78.9/100	Adequate / Neutral
Epistemic Grounding	89.5/100	Excellent / Restrained
Punctuation Discipline	87.2/100	Excellent / Restrained
Readability Structure	89.5/100	Excellent / Restrained
Lexical Diversity	79.9/100	Adequate / Neutral

OVERALL CREDIBILITY INDEX
85.7 / 100
Credibility Tier: A (Restrained / Professional Copy)

Scores are standardized against verified newsroom copy standards: high factual attribution, balanced syntax, and low sensationalism maximize the index.

3. Grounded Diagnostic Reasoning & Counterfactual Sensitivity

Executive Synthesis: Verdict is UNCERTAIN: No matching authoritative evidence exists in the knowledge base to verify or debunk the claims. Per trustworthy AI policy, the system abstains rather than forcing an ungrounded guess.

Diagnostic Explanation: 1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **95.9% calibrated confidence**.
- **Gating Trigger**: System abstention triggered under safety policy (missing_evidence).

Counterfactual Simulation: Neutralizing 1 linguistic markers resulted in a signed probability shift of -0.0 percentage points (from 95.9% to 95.9%), retaining the primary verdict (FAKE). Original P(FAKE): **95.9%** → Transformed P(FAKE): **95.9%** **(Signed Shift: -0.0%)**

[v] Clean Stylometrics: No significant sensational exaggeration, clickbait patterns, or manipulative phrasing detected.

4. Claim-Level Semantic Evidence Grounding (Triage Verification)

Extracted Factual Claim	Verification Status	Authoritative Source / Match	Evidence Context Quote
A special Pocso court in Agra has sentenced a 30-year-old man to life imprisonment...	UNVERIFIED (Sim: 0.03)	No verified source found	No corroborating evidence located in authoritative knowledge base or retrieved sou...
According to prosecution, on the night of April 17, 2025, the girl's father went o...	UNVERIFIED (Sim: 0.11)	No verified source found	No corroborating evidence located in authoritative knowledge base or retrieved sou...
During the trial, the prosecution examined 12 witnesses, including the complainant...	UNVERIFIED (Sim: 0.17)	No verified source found	No corroborating evidence located in authoritative knowledge base or retrieved sou...
The court also imposed a fine of Rs 50,000 on him.	UNVERIFIED (Sim: 0.04)	No verified source found	No corroborating evidence located in authoritative knowledge base or retrieved sou...

5. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Impact	Top REAL-Driving Anchors	Impact
Logistic Regression [P(FAKE)]	+1.946	Base Disagreement (std)	-0.205
LightGBM [P(FAKE)]	+0.582	-	-
Random Forest [P(FAKE)]	+0.494	-	-

6. Empirical Benchmark Model Comparison (Held-out Test Split)

Model Name	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest (Uncalibrated)	0.8557	0.8552	0.1488	100.0%	0.8557
XGBoost (Uncalibrated)	0.8705	0.8703	0.0371	100.0%	0.8705
LightGBM (Uncalibrated)	0.8787	0.8785	0.0401	100.0%	0.8787
Logistic Regression (Uncalibrated)	0.8951	0.8951	0.0372	100.0%	0.8951
Uncalibrated Stack	0.8984	0.8983	0.0372	100.0%	0.8984
Calibrated Stack (Full Coverage)	0.8967	0.8967	0.0276	100.0%	0.8967
CAGE-Fake (Calibrated + Gated)	**0.8967**	**0.8967**	**0.0276**	**84.6%**	**0.9496**

7. Research Calibration & Disagreement Visualizations







